

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278374

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Norberg; Robert et al.

I/O Station Storing Diagnostics Data

Abstract

The present disclosure relates to an I/O station, connected with a plurality of network devices in an automation system. The I/O station includes a plurality of I/O modules, each I/O module providing one or more I/O channels, each I/O channel connecting a respective field device. The I/O station also includes an I/O network interface for forwarding data between the I/O modules and the network devices via a network. The I/O station also includes a data storage storing a set of diagnostics data, the set including a plurality of entries of diagnostics data, each entry relating to performance of the automation system. The I/O station is operative to, in response to a request from anyone of the network devices, send the stored set of diagnostics data to the requesting network device.

Inventors: Norberg; Robert (Eskilstuna, SE), Pook; Stefan (Minden, DE), Åslund; Jörgen (Västerås, SE), Viero; Enrico (Västerås, SE), Hansson; Ulf (Västerås, SE)

Applicant: ABB Schweiz AG (Baden, CH)

Family ID: 90123387

Appl. No.: 19/056973

Filed: February 19, 2025

Foreign Application Priority Data

EP	24160542.7	Feb. 29, 2024
----	------------	---------------

Publication Classification

Int. Cl.: G06F13/20 (20060101)

U.S. Cl.:

CPC G06F13/20 (20130101); G06F2213/40 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an input-output (I/O) station, connected with a plurality of network devices in an industrial automation system.

BACKGROUND

[0002] An I/O station, also called I/O system, is used in an industrial automation system for connecting field devices, such as sensors, actuators or the like in an industrial plant to controllers of the plant. The I/O station typically comprises a Fieldbus Communication Interface (FCI) and I/O modules connecting the field devices, where the FCI interface is arranged for forwarding data between the modules and the controllers, via fieldbus or fieldbus network. Since the controllers are digital while at least some of the field devices may be analog legacy devices (e.g. requiring a 4-20 mA signal), the I/O station may convert analogue signals to digital and vice versa. Each of the I/O modules may be any of a digital-in (DI) module, digital-out (DO) module, analog-in (AI) module or analog-out (AO) module, or a combination thereof, or configurable as in/out analog or configurable as in/out digital or configurable as in/out analog/digital.

[0003] Diagnostics related to wiring or field devices are present in many automation systems today. In most cases this diagnostics is interrupt driven on the fieldbus. If an I/O module has something to report, it informs the controller that it holds diagnostics and the controller goes out and fetches this data, normally via acyclic services. The controller will then distribute this diagnostics information to alarm and event list of the HMI and the engineering tool.

[0004] Modern automation systems are getting more accessible due to introduction of I/O networks and there is an increasing market need for more service and asset management tools/applications collecting data including diagnostics for preventive maintenance and other purposes. A normal solution today is that these tools pick up the data and diagnostics indirectly via the alarm and event lists and engineering tools.

SUMMARY

[0005] It is an objective of the present invention to provide an improved I/O station which can provide diagnostics data to the network.

[0006] According to an aspect of the present invention, there is provided an I/O station, connected with a plurality of network devices in an automation system. The I/O station comprises a plurality of I/O modules, each I/O module providing one or more I/O channels, each I/O channel connecting a respective field device. The I/O station also comprises an I/O network interface for forwarding data between the I/O modules and the network devices via a network (3). The I/O station also comprises a data storage storing a set of diagnostics data, the set comprising a plurality of entries of diagnostics data, each entry relating to performance of the automation system. The I/O station is operative to, in response to a request from anyone of the network devices, send the stored set of diagnostics data to the requesting network device. Embodiments of the I/O station of the present disclosure may be configured to perform an embodiment of the method of the present disclosure. Preferably, at least one of the entries in the stored set of diagnostics data relates to a condition of one of the I/O modules before (e.g. just before) a shutdown of said I/O module, typically an unscheduled shutdown.

[0007] According to another aspect of the present invention, there is provided an automation system comprising a plurality of network devices, a plurality of the I/O stations of the present disclosure, and the network via which the I/O stations are communicatively connected to the network devices.

[0008] According to another aspect of the present invention, there is provided a method performed by an embodiment of the I/O station of the present disclosure. The method comprises obtaining

diagnostics data relating to performance of the automation system. The method also comprises storing the obtained diagnostics data as an entry in the set of diagnostics data in the data storage. The method also comprises, from one of the network devices, receiving a request for the set of diagnostics data. The method also comprises, in response to the received request, sending the set of diagnostics data, including the stored entry, to said one of the network devices.

[0009] According to another aspect of the present invention, there is provided a computer program product comprising computer-executable components for causing an I/O station to perform an embodiment of the method of the present disclosure when the computer-executable components are run on processing circuitry comprised in the I/O station.

[0010] By the I/O station storing a set of diagnostics data, the I/O station can provide the whole set of diagnostics data to a network device in response to a request from said network device. Thus, collection of diagnostics data does not have to be only interrupt driven. The I/O station may also hold a local copy of all diagnostics and information related to the I/O station, wiring and field devices. This local copy may then be collected by any network device (including service and asset management tools/applications) directly from the I/O station, e.g. via specific fieldbus commands. This leads to easier access to data and diagnostics in the automation system.

[0011] It is to be noted that any feature of any of the aspects may be applied to any other aspect, wherever appropriate. Likewise, any advantage of any of the aspects may apply to any of the other aspects. Other objectives, features and advantages of the enclosed embodiments will be apparent from the following detailed disclosure, from the attached dependent claims as well as from the drawings.

[0012] Generally, all terms used in the claims are to be interpreted according to their ordinary meaning in the technical field, unless explicitly defined otherwise herein. All references to “a/an/the element, apparatus, component, means, step, etc.” are to be interpreted openly as referring to at least one instance of the element, apparatus, component, means, step, etc., unless explicitly stated otherwise. The steps of any method disclosed herein do not have to be performed in the exact order disclosed, unless explicitly stated. The use of “first”, “second” etc. for different features/components of the present disclosure are only intended to distinguish the features/components from other similar features/components and not to impart any order or hierarchy to the features/components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Embodiments will be described, by way of example, with reference to the accompanying drawings, in which:

[0014] FIG. 1 is a schematic block diagram of an automation system, in accordance with some embodiments of the present invention.

[0015] FIG. 2 is a schematic block diagram of an I/O station, in accordance with some embodiments of the present invention.

[0016] FIG. 3 is a schematic block diagram of an I/O station comprising processing circuitry and a data storage, in accordance with some embodiments of the present invention.

[0017] FIG. 4 is a schematic circuit diagram of an I/O module, in accordance with some embodiments of the present invention.

[0018] FIG. 5 is a schematic flow chart of some embodiments of the method of the present invention.

DETAILED DESCRIPTION

[0019] Embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which certain embodiments are shown. However, other embodiments

in many different forms are possible within the scope of the present disclosure. Rather, the following embodiments are provided by way of example so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Like numbers refer to like elements throughout the description.

[0020] FIG. 1 illustrates an automation system **1** comprising a plurality of I/O stations **10**. An I/O station may be embedded hardware and software that, for inputs converts the physical signal from the process running in the plant (either directly from the process or via a field device) to digital values, and for outputs converts digital values into physical signals for the process (either directly to the process or via a field device). The I/O stations **10** are connected with a plurality of network devices via a network **3**. The network may comprise one or more fieldbus(es). The network **3** may be configured for communication in accordance with any fieldbus protocol, e.g. an OPC Unified Architecture, OPC UA, or a Process Field Network, PROFINET, communication protocol, preferably OPC UA.

[0021] The network devices may comprise any of one or more controller(s) **2**, Human-Machine Interface(s) (HMI) **5**, engineering tool(s) **6** and/or any other device **7** (typically processing device) running an application (e.g. a service application) e.g. for asset management in the plant comprising the automation system **1**. For instance, the network devices connected in the network **3** may comprise a plurality of controller(s) **2**, one or more, e.g. only one, HMI **5** and/or one or more, e.g. only one, engineering tool **6**. A controller **2** may be embedded hardware and software that executes a real-time plant process specific application that e.g. reads signals from the I/O stations **10**, writes signals to the I/O stations **10** and/or receives commands from an HMI **5**. An HMI **5** may convene to an operator the status of the process based on reading data from the controller(s) **2** and/or the I/O stations **10**. The HMI **5** may additionally command a controller **2** to start an action. An engineering tool **6** may configure the binding and conversion of I/O channels of an I/O station **10** and/or the application in a controller **2**. The engineering tool **6** may execute on a PC where activities are not time critical.

[0022] In the example of FIG. 1, there are three I/O stations **10** and three controllers **2**. However, there need not be a one-to-one relationship between the number of I/O stations and the number of controllers and any one of the controllers may connect with any of the I/O channels in any of the I/O stations. In some embodiments, at least one of the network devices **2**, **5**, **7** and/or **6**, e.g. at least one controller **2**, may be communicatively connected (i.e. connected such that it is able to communicate, e.g. wirelessly or non-wirelessly) with I/O channels of at least two of the I/O stations **10**.

[0023] In the example of FIG. 1, the controllers **2** are connected between the I/O stations **10** and the HMI **5**. In other examples, the HMI **5** may be connected directly to the stations **10**, without being connected via a controller. Also, other topologies are possible. The engineering tool **6** may be connected to any or all of the I/O stations **10**, and/or to any of the controllers **2**, e.g. via the network **3**. Typically, the engineering tool **6** is only connected when needed, but in some embodiments, it may be permanently connected.

[0024] Each of the I/O stations **10** comprises a plurality of I/O modules **12** which are interfaced with the network **3** via an I/O network interface **11**. The modules **12** may be connected to the network interface **11** via one or more I/O bus(es) **13** within the station **10**, and/or there may be point-to-point connections between the modules **12** and network interface **11**.

[0025] FIG. 2 illustrates an I/O station **10**, e.g. one of the stations **10** in FIG. 1, connecting a plurality of field devices **21**. Each I/O module **12** provides one or more I/O channel(s) **20**, where each channel **20** can connect a respective field device **21** (i.e. one field device per channel and one channel per field device). In an example, one of the I/O modules **12** provides only one channel **20** and may thus be able to connect only one field device **21**. In another example, one of the I/O modules **12** provides **16** channels **20**, enabling the module to connect up to **16** field devices **21**. There may be unused channels **20** (i.e. channels to which no field device is connected), but each of

the channels **20** which connects a field device **21** is communicatively connected to at least one of the network devices, e.g. one or more controllers **2**, via the network interface **11** and the network **3**. [0026] The modules **12** and/or the network interface **11** may be mechanically connected to one or more baseplate(s) **26** within the station **10**. In some embodiment a single baseplate **26** provides slots for mechanically connecting the modules **12** and the network interface **11**. In other embodiments, that functionality is provided by a plurality of modular baseplates **26**, one of which providing a slot for mechanically connecting the network interface **11**.

[0027] The station **10** comprises a data storage **22**, typically a non-volatile data storage or a combination of volatile and non-volatile data storage, storing a set of diagnostics data. Typically, the set of diagnostics data comprises a plurality of entries of diagnostics data, each entry relating to performance of the automation system **1**, typically to a part of the automation system **1** associated with the I/O station **10** comprising the data storage **22**, e.g. to performance of the I/O network interface **11** comprised in the I/O station, of one of the field devices **21** connected to the I/O station, of one of the I/O modules **12** comprised in the I/O station or of one of the I/O channels **20**, e.g. wiring thereof, provided by an I/O module comprised in the I/O station. In some embodiments, at least a part of the storage **22** (e.g. holding at least one of the entries in the set of set of diagnostics data **23**) is comprised in the network interface **11**. In some embodiments, at least part of the storage **22** (e.g. holding at least one of the entries in the set of set of diagnostics data **23**) is comprised in the baseplate **26**, e.g. in a baseplate mechanically connecting the network interface **11**. In some embodiments, at least part of the storage **22** (e.g. holding at least one of the entries in the set of set of diagnostics data **23**) is comprised in at least one of the I/O modules **12**. For instance, the data storage **22** may comprise non-volatile storage in the network interface **11** or base plate **26**, as well as volatile and/or non-volatile storage in each of the I/O modules **12**.

[0028] FIG. **3** schematically illustrates an embodiment of an I/O station **10** of the present disclosure. The I/O station **10** comprises processing circuitry **25** e.g. a central processing unit (CPU). The processing circuitry **25** may comprise one or a plurality of processing units in the form of microprocessor(s). However, other suitable devices with computing capabilities could be comprised in the processing circuitry **25**, e.g. an application specific integrated circuit (ASIC), a field programmable gate array (FPGA),

[0029] System on Chip (SoC) or a complex programmable logic device (CPLD). The processing circuitry **25** is configured to run one or several computer program(s) or software (SW) **24** stored in a storage **22** of one or several storage unit(s) e.g. a memory. The storage unit is regarded as a computer readable means **22**, forming a computer program product together with the SW **24** stored thereon as computer-executable components, and may e.g. fully or partly be in the form of a Random Access Memory (RAM), a Flash memory or other solid-state memory, or a hard disk, or be a combination thereof. The processing circuitry **25** may also be configured to store data in the storage **22**, as needed. The data storage **22** also comprises, i.e. holds, the set of diagnostics data **23** discussed herein.

[0030] FIG. **4** illustrates a first I/O module **12a** providing at least one I/O channel **20** to a respective field device **21**. For redundancy, a second I/O module **12b** may provide an I/O channel **20** to the same field device **21**. For instance, in case the first I/O module **12a** breaks, the second I/O module **12b** can take over. The baseplate **26** may route the wiring of the channel **20** to both modules **12a** and **12b**.

[0031] The I/O module **12a**, or each of the I/O modules **12**, may comprise at least a part of the storage **22**, e.g. holding at least one of the entries in the set of set of diagnostics data **23** in said module. The storage **22** in the module **12** may comprise or consist of at least one non-volatile storage (e.g. memory) **41** and/or at least one volatile storage (e.g. memory) **42**. A microprocessor (UP) or other processing circuitry **46** of the module **12** may be configured to store entries in the set of diagnostics data said non-volatile memory **41** and/or volatile memory **42** in the module **12**. Entries stored in parts **41** and/or **42** of the data storage **22** in the module **12** may then be

communicated via e.g. the I/O bus **13** for sending the stored set of diagnostics data **23** to a requesting network device.

[0032] Examples of diagnostics data **23** which may be stored as entries of the set in the data storage **22** are schematically illustrated in FIG. **4**. For instance, a line sensor **44** may monitor the performance (e.g. health) of the channel **20** of the field device **21**, e.g. of the channel **20** between the field device **21** and an Analog-to-Digital (A/D) converter **45** in the module **12**. If it is determined that the channel is not adequately operational, e.g. due to a cable break, the LS **44** may send a sensor signal to the microprocessor **46**, said sensor signal including information about the channel performance, and the microprocessor may then save diagnostics data relating to the channel performance as an entry of the set of diagnostics data in the data storage **22**, e.g. in the non-volatile storage **41** or the volatile storage **42**, preferably the volatile storage **42** which may be more suited for storing real-time diagnostics data such as from the LS **44**.

[0033] Another example relates to monitoring of a reference voltage U_{ref} (e.g. 2 V) by a reference generator **47** to an A/D converter **45**. A transformer **43** is used for transforming an incoming voltage U_{in} fed to the module **12** (e.g. 24 V) to internal voltages U_1 and U_2 (e.g. 5 V and 12 V, respectively) used within the module **12**. In the example of FIG. **4**, the reference generator **47** is fed U_2 and generates U_{ref} which is provided to the A/D converter **45**. A sensor may monitor the operation of the reference generator **47**, e.g. monitor that the reference voltage U_{ref} is correct and stable, and may send a sensor signal to the microprocessor **46**, said sensor signal including information about the performance of the reference generator **47**, and the microprocessor may then save diagnostics data relating to said performance as an entry of the set of diagnostics data in the data storage **22**, e.g. in the non-volatile storage **41** or the volatile storage **42**, preferably the volatile storage **42** which may be more suited for storing real-time diagnostics data.

[0034] FIG. **5** is a schematic flow chart illustrating some embodiments of a method of the present invention. The method is performed by an I/O station **10**, e.g. as discussed herein. The method comprises obtaining S_1 diagnostics data **23**, e.g. relating to performance of one of the field devices **21**, one of the I/O modules **12** or one of the I/O channels **20**. The method also comprises storing S_2 the obtained S_1 diagnostics data as an entry in the set of diagnostics data **23** in the data storage **22**. Typically, the entry being stored (i.e. saved) may replace a corresponding older entry in the set. Thus, the set of diagnostics data is always up to date with the latest available diagnostics data and space is not taken up by out-of-date diagnostics data. As discussed herein, the data storage **22** may comprise several different parts in different parts of the I/O station **10**, and may comprise volatile and/or non-volatile parts **42** and/or **41**. The method also comprises receiving S_3 , from one of the network devices, a request for the set of diagnostics data **23**. The method also comprises sending S_4 , in response to the received S_3 request, the set of diagnostics data **23**, including the stored S_2 entry, to said one of the network devices. The set may for instance be sent S_4 in a message addressed to the requesting network device, i.e. not broadcast. For instance, the network device may send a specific commando, corresponding to “send all diagnostics”, whereby the I/O station **10**, in response thereto, sends the whole set of entries to said network device. This is in contrast with situations of prior art where the I/O station sends only specific fault information on its own volition when a fault is detected.

[0035] In some embodiments of the present invention, the diagnostics data which is stored S_2 as the entry in the set of diagnostics data **23** replaces previously stored diagnostics data in the set, thus updating the set of diagnostics data. The replaced entry, including the diagnostics data of said replaced entry, may typically be automatically deleted from the storage **22** as part of the storing (i.e. saving) S_2 of the new entry of diagnostics data.

[0036] In some embodiments of the present invention, the storing S_2 of the entry comprises associating a time stamp with the stored entry, the timestamp indicating a time when the diagnostics data of the entry was obtained S_1 . Thus, at least one of the entries in the set of diagnostics data **23**, e.g. each of some or all of the entries in the set of diagnostics data, is stored

(i.e. held) in the storage **22** in association with a respective timestamp indicating a time when the diagnostics data of the entry was obtained. Having respective timestamps associated with the entries enable analysis of the chain of events resulting in the diagnostics data in the set. Thus, it may be possible to determine from the diagnostics data in the set in which order different things happened/is happening in the I/O station **10**, possibly enabling determining causality.

[0037] In some embodiments of the present invention, at least one of the entries in the stored set of diagnostics data **23** relates to a condition of one of the I/O modules **12** before shutdown of said I/O module, e.g. due to detection of an internal error in the module. Such an entry may e.g. be stored in a non-volatile storage **41** in said module **12**. By means of such an entry with diagnostic data relating to a condition of the module before shutdown, analysis to determine the cause of the shutdown may be facilitated. This could be regarded as a post mortem analysis. Thus, especially in case of an unscheduled shutdown, or other irregularity, of the module, diagnostic data relating to the condition of the module just before the shutdown/irregularity, e.g. at the time of shutdown or at a time before the shutdown closest to the time of shutdown for which diagnostic data is available, may be stored as an entry in the set of diagnostics data.

[0038] In some embodiments of the present invention, the I/O station **10**, e.g. the I/O network interface **11**, is configured to send the stored set of diagnostics data **23** via the network **3** in accordance with an OPC UA or PROFINET communication protocol. Similarly, it may be convenient to store the diagnostics data in the data storage **22** in a format which is compatible with the OPC UA or PROFINET communication protocol.

[0039] The present disclosure has mainly been described above with reference to a few embodiments. However, as is readily appreciated by a person skilled in the art, other embodiments than the ones disclosed above are equally possible within the scope of the present disclosure, as defined by the appended claims.

Claims

1. An input-output, I/O, station, connected with a plurality of network devices in an automation system, the I/O station comprising: a plurality of I/O modules, each I/O module providing one or more I/O channels, each I/O channel connecting a respective field device; an I/O network interface for forwarding data between the I/O modules and the network devices via a network; a data storage storing a set of diagnostics data, the set comprising a plurality of entries of diagnostics data, each entry relating to performance of the automation system; wherein the I/O station is operative to, in response to a request from anyone of the network devices, send the stored set of diagnostics data to the requesting network device; and wherein at least one of the entries in the stored set of diagnostics data relates to a condition of one of the I/O modules just before an unscheduled shutdown of said I/O module.
2. The I/O station of claim 1, wherein each of the entries relating to performance of the automation system relates to performance of the network interface, one of the field devices, one of the I/O modules or one of the I/O channels of the I/O station.
3. The I/O station of claim 1, wherein each of some or all of the entries in the stored set of diagnostics data is stored in association with a respective timestamp indicating a time when the diagnostics data of the entry was obtained.
4. The I/O station of claim 1, wherein the data storage is comprised in the I/O network interface and/or in one or more of the I/O modules and/or in a baseplate providing a slot to which the I/O network interface and/or I/O module is mechanically connected in the I/O station.
5. The I/O station of claim 1, wherein the network devices include any of: one or more controller(s), one or more human-machine interface(s) and one or more engineering tool(s), and any other device, e.g. a device running a service application, connected to the network.
6. The I/O station of claim 1, wherein the I/O station, e.g. the I/O network interface, is configured

to send the stored set of diagnostics data via the network in accordance with an OPC Unified Architecture, OPC UA, or Process Field Network, PROFINET, communication protocol.

7. An automation system comprising: a plurality of network devices; a plurality of the I/O stations connected with the plurality of network devices, each I/O station including: a plurality of I/O modules, each I/O module providing one or more I/O channels, each I/O channel connecting a respective field device; an I/O network interface for forwarding data between the I/O modules and the network devices via a network; a data storage storing a set of diagnostics data, the set comprising a plurality of entries of diagnostics data, each entry relating to performance of the automation system; wherein the I/O station is operative to, in response to a request from anyone of the network devices, send the stored set of diagnostics data to the requesting network device; wherein at least one of the entries in the stored set of diagnostics data relates to a condition of one of the I/O modules just before an unscheduled shutdown of said I/O module; and the network via which the I/O stations are communicatively connected to the network devices.

8. A method performed by an input-output, I/O, station, connected with a plurality of network devices in an automation system, the I/O station including: a plurality of I/O modules, each I/O module providing one or more I/O channels, each I/O channel connecting a respective field device; an I/O network interface for forwarding data between the I/O modules and the network devices via a network; a data storage storing a set of diagnostics data, the set comprising a plurality of entries of diagnostics data, each entry relating to performance of the automation system; wherein the I/O station is operative to, in response to a request from anyone of the network devices, send the stored set of diagnostics data to the requesting network device; wherein at least one of the entries in the stored set of diagnostics data relates to a condition of one of the I/O modules just before an unscheduled shutdown of said I/O module, the method comprising: obtaining diagnostics data relating to performance of the automation system; storing the obtained diagnostics data as an entry in the set of diagnostics data in the data storage; from one of the network devices, receiving a request for the set of diagnostics data; and in response to the received request, sending the set of diagnostics data, including the stored entry, to said one of the network devices.

9. The method of claim 8, wherein the diagnostics data which is stored as the entry in the set of diagnostics data replaces previously stored diagnostics data in the set, thus updating the set of diagnostics data.

10. The method of claim 8, wherein the storing of the entry includes associating a timestamp with the stored entry, the timestamp indicating a time when the diagnostics data of the entry was obtained.

11. A computer program product comprising computer-executable components for causing an I/O station connected with a plurality of network devices in an automation system, the I/O station including: a plurality of I/O modules, each I/O module providing one or more I/O channels, each I/O channel connecting a respective field device, an I/O network interface for forwarding data between the I/O modules and the network devices via a network, a data storage storing a set of diagnostics data, the set comprising a plurality of entries of diagnostics data, each entry relating to performance of the automation system, wherein the I/O station is operative to, in response to a request from anyone of the network devices, send the stored set of diagnostics data to the requesting network device, and wherein at least one of the entries in the stored set of diagnostics data relates to a condition of one of the I/O modules just before an unscheduled shutdown of said I/O module; to perform a method comprising: obtaining diagnostics data relating to performance of the automation system, storing the obtained diagnostics data as an entry in the set of diagnostics data in the data storage, from one of the network devices, receiving a request for the set of diagnostics data, and in response to the received request, sending the set of diagnostics data, including the stored entry, to said one of the network devices when the computer-executable components are run on processing circuitry comprised in the I/O station.

12. The I/O station of claim 2, wherein each of some or all of the entries in the stored set of

diagnostics data is stored in association with a respective timestamp indicating a time when the diagnostics data of the entry was obtained.

13. The I/O station of claim 2, wherein the data storage is comprised in the I/O network interface and/or in one or more of the I/O modules and/or in a baseplate providing a slot to which the I/O network interface and/or I/O module is mechanically connected in the I/O station.

14. The I/O station of claim 2, wherein the network devices include any of: one or more controller(s), one or more human-machine interface(s) and one or more engineering tool(s), and any other device, e.g. a device running a service application, connected to the network.

15. The I/O station of claim 2, wherein the I/O station, e.g. the I/O network interface, is configured to send the stored set of diagnostics data via the network in accordance with an OPC Unified Architecture, OPC UA, or Process Field Network, PROFINET, communication protocol.
